

Credit Card Fraud Detection using Deep Learning

Problem statement

Credit card fraud is a growing financial threat, impacting tens of millions of Americans each year. Detecting fraudulent transactions is challenging due to extreme class imbalance, evolving fraud patterns, and the high cost of false negatives. Fraud detection is dominated by classical machine learning algorithms, with there being limited research using deep learning. This project aims to investigate the effectiveness of an ensemble neural network approach, combining a feed forward neural network and an Autoencoder.

Motivation:

Global credit card and debit card fraud losses reached approximately \$34 billion to \$35 billion annually as of 2022–2023. The United States accounts for over 40% of these global losses. Fraud detection directly protects consumers from financial losses, and in the real world, fast and accurate fraud detection is crucial for financial transactions.

Related Works:

[Pumsirirat, Apapan. "Credit card fraud detection using deep learning based on auto-encoder and restricted boltzmann machine." International Journal of advanced computer science and applications \(2018\)](#)

[Alarfaj, Fawaz Khaled, et al. "Credit card fraud detection using state-of-the-art machine learning and deep learning algorithms." Ieee Access 10 \(2022\): 39700-39715.](#)

These papers look at the effectiveness of a CNN and an autoencoder respectively. Our project proposes an ensemble approach combining both models and evaluating whether it increases the effectiveness of detection.

Method(s):

Proposed Dataset:

- [Credit Card Fraud Detection Kaggle](#)
- Normal transactions (~284,315)
- Fraudulent transactions (~492)
- It contains only numerical input variables which are the result of a PCA transformation. This means that direct interpretation of the variables will be difficult.
- the only features which have not been transformed with PCA are 'Time' and 'Amount'.
- 'Time' contains the seconds elapsed between each transaction and the first transaction in the dataset.
- 'Amount' is the transaction Amount.
- 'Class' is the response variable and it takes value 1 in case of fraud and 0 otherwise.

Another option for the dataset:

- [BankSim Synthetic Payments Simulator Dataset](#)
- Synthetic data based on a sample of aggregated transactional data provided by a bank in Spain
- 594643 records (587443 are normal payments and 7200 fraudulent)

- Would require more EDA than the first dataset, but would be more interpretable due to the data not being pre-transformed through PCA.

FFN Classifier:

- Train a feedforward neural network with weighted/focal loss to handle class imbalance.
- Output probability of fraud for each transaction.
- Binary classification problem (fraud or not fraud)
- FFN is trained on the full dataset (Need true and predicted value)

Autoencoder (Anomaly Detection):

- Train an autoencoder on normal transactions.
- Calculate reconstruction error as an anomaly score; convert to probability.
- Only trained on the non-fraud transactions

Ensemble Method:

- Combine FFN probability and Autoencoder anomaly score via weighted average or stacking.
- Threshold ensemble output to classify fraud/non-fraud.

Evaluation: Use normal metrics such as:

- Metrics: PR-AUC, F1-score, Precision, Recall.
- Compare FFN alone, Autoencoder alone, ensemble
- *POSSIBLE EXTRA IDEA*
- Train classical ML models such as Logistic Regression, Random Forest for comparison.

Team members:

Timetable:

Be sure to emphasize the team efforts achieving the deadlines.